he had many regrets; it makes perfect sense to list the most important categories of either; but it is a non-question to ask *how many*.'"

Alice's father didn't really seem satisfied by this. After a thoughtful pause, he responded, "Look, I'm trying to be fair here. I'll accept that the worlds are not fundamental, so there is something approximate about how they are defined. But surely you can tell me whether there are just a finite number of them or the number is truly infinite."

"It's a fair question," Alice agreed, maybe a bit reluctantly. "Unfortunately, we don't know the answer. There's an upper limit to the number of worlds, which is just the size of Hilbert space, the space of all possible wave functions."

"But we know that Hilbert space is infinitely big," interjected her father. "Even for just one particle, Hilbert space is infinite-dimensional, not to mention for quantum field theory. So the number of worlds sounds like it's infinite."

"We're not sure whether the Hilbert space for our actual universe has a finite or infinite number of dimensions. We certainly know of some systems for which the appropriate Hilbert space is finite-dimensional. A single qubit is either spin-up or spin-down, so it corresponds to a two-dimensional Hilbert space. If we have N qubits, the corresponding Hilbert space is 2^N-dimensional—the size of Hilbert space grows exponentially as we include more particles. A cup of coffee contains roughly 10^{25} electrons, protons, and neutrons, each of whose spins is described by a qubit. So the Hilbert space for a cup of coffee—just including the spins, not yet worrying about the locations of the particles—has a dimensionality of about $2^{10^{25}}$.

"Needless to say," continued Alice, "that's a crazy-big number. One followed by 10^{25} zeroes, if you wrote it in binary. Which you wouldn't have time to do, even if you had been working for the entire lifetime of our observable universe."

"But you're obviously cheating, the real number is much bigger than that," said her father. "You're counting spins, but real particles have locations in space too. And there are an infinite number of such locations. That's why the Hilbert space for a collection of particles is infinite-dimensional—the number of dimensions is just the number of possible measurement outcomes."

"Right. And it's true, Hugh Everett himself thought that every quantum measurement split the universe into an infinite number of worlds, and he